Paper/Subject B.E/B.Tech.(Paper 1)
Exam Date 29 Jul 2022
Exam Slot 1

Question ID:100201
Topic Name: Mathematics-Section A
Question:

Let R be a relation from the set {1, 2, 3, . . . , 60} to itself such that $R = \{(a, b) : b = pq$, where $p, q \geq 3$ are prime numbers$\}$. Then, the number of elements in R is :

A 600

B 660

C 540

D 720

Answer Given By Candidate:**C**

Question ID:100202
Topic Name: Mathematics-Section A
Question: If $z = 2 + 3i$, then $z^5 + (\overline{z})^5$ is equal to :

A 244

B 224

C 245

D 265

Answer Given By Candidate:**Not Attempted**

Question ID:100203
Topic Name: Mathematics-Section A
Question:

Let A and B be two 3×3 non-zero real matrices such that AB is a zero matrix. Then

A the system of linear equations $AX = 0$ has a unique solution

B the system of linear equations $AX = 0$ has infinitely many solutions

C B is an invertible matrix

D adj(A) is an invertible matrix

Answer Given By Candidate:**Not Attempted**

Question ID:100204
Topic Name: Mathematics-Section A

Question:

If $\frac{1}{(20-a)(40-a)}+\frac{1}{(40-a)(60-a)}+\ldots+\frac{1}{(180-a)(200-a)}=\frac{1}{256}$, then the maximum value of a is :

A 198

B 202

C 212

D 218

Answer Given By Candidate:**Not Attempted**

Question ID:100205
Topic Name: Mathematics-Section A
Question:

If $\lim_{x\to 0}\frac{\alpha e^{x}+\beta e^{-x}+\gamma \sin x}{x\sin^{2}x}=\frac{2}{3}$, where $\alpha, \beta, \gamma \in \mathbf{R}$, then which of the following is **NOT** correct ?

A $\alpha^2+\beta^2+\gamma^2=6$

B $\alpha\beta+\beta\gamma+\gamma\alpha+1=0$

C $\alpha\beta^2+\beta\gamma^2+\gamma\alpha^2+3=0$

D $\alpha^2-\beta^2+\gamma^2=4$

Answer Given By Candidate:**Not Attempted**

Question ID:100206
Topic Name: Mathematics-Section A

Question:

The integral $\int_{0}^{\frac{\pi}{2}}\frac{1}{3+2\sin x+\cos x}\,dx$ is equal to :

A $\tan^{-1}(2)$

B $\tan^{-1}(2)-\frac{\pi}{4}$

C $\frac{1}{2}\tan^{-1}(2)-\frac{\pi}{8}$

D $\frac{1}{2}$

Answer Given By Candidate:**Not Attempted**

Question ID:100207
Topic Name: Mathematics-Section A

Question:

Let the solution curve $y=y(x)$ of the differential equation $\left(1+e^{2x}\right)\left(\frac{dy}{dx}+y\right)=1$ pass through the point $\left(0, \frac{\pi}{2}\right)$. Then, $\lim\limits_{x\to\infty} e^{x}y(x)$ is equal to :

A $\frac{\pi}{4}$

B $\frac{3\pi}{4}$

C $\frac{\pi}{2}$

D $\frac{3\pi}{2}$

Answer Given By Candidate:**Not Attempted**

Question ID:100208
Topic Name: Mathematics-Section A
Question:

Let a line L pass through the point of intersection of the lines $bx+10y-8=0$ and $2x-3y=0$, $b \in \mathbf{R}-\left\{\frac{4}{3}\right\}$. If the line L also passes through the point (1, 1) and touches the circle $17(x^2+y^2)=16$, then the eccentricity of the ellipse $\frac{x^2}{5}+\frac{y^2}{b^2}=1$ is :

A $\frac{2}{\sqrt{5}}$

B $\sqrt{\frac{3}{5}}$

C $\frac{1}{\sqrt{5}}$

D $\sqrt{\frac{2}{5}}$

Answer Given By Candidate:**Not Attempted**

Question ID:100209
Topic Name: Mathematics-Section A
Question:

If the foot of the perpendicular from the point $A(-1, 4, 3)$ on the plane $P: 2x+my+nz=4$, is $\left(-2, \frac{7}{2}, \frac{3}{2}\right)$, then the distance of the point A from the plane P, measured parallel to a line with direction ratios 3, -1, -4, is equal to :

A 1

B $\sqrt{26}$

C $2\sqrt{2}$

D $\sqrt{14}$

Answer Given By Candidate:**Not Attempted**

Question ID:100210
Topic Name: Mathematics-Section A
Question:

Let $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda \vec{c}$. If $\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is :

A -5

B 5

C 1

D -1

Answer Given By Candidate:**Not Attempted**

Question ID:100211
Topic Name: Mathematics-Section A
Question:

The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is $\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then $\cot\alpha$ is equal to :

A $\frac{6}{5}$

B $\frac{9}{5}$

C $\frac{4}{3}$

D $\frac{7}{3}$

Answer Given By Candidate:**Not Attempted**

Question ID:100212
Topic Name: Mathematics-Section A
Question: The statement $(p \wedge q) \Rightarrow (p \wedge r)$ is equivalent to :

A $q \Rightarrow (p \wedge r)$

B $p \Rightarrow (p \wedge r)$

C $(p \wedge r) \Rightarrow (p \wedge q)$

D $(p \wedge q) \Rightarrow r$

Answer Given By Candidate:**C**

Question ID:100213
Topic Name: Mathematics-Section A
Question:

Let the circumcentre of a triangle with vertices A(a, 3), B(b, 5) and C(a, b), ab > 0 be P(1, 1). If the line AP intersects the line BC at the point $Q(k_1, k_2)$, then $k_1 + k_2$ is equal to :

A 2

B $\frac{4}{7}$

C $\frac{2}{7}$

D 4

Answer Given By Candidate:**Not Attempted**

Question ID:100214
Topic Name: Mathematics-Section A
Question:

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $(\hat{a} + \hat{b})$ and $(\hat{a} + 2\hat{b} + 2(\hat{a} \times \hat{b}))$, then the value of $164\cos^2\theta$ is equal to :

A $90 + 27\sqrt{2}$

B $45 + 18\sqrt{2}$

C $90 + 3\sqrt{2}$

D $54 + 90\sqrt{2}$

Answer Given By Candidate:**Not Attempted**

Question ID:100215
Topic Name: Mathematics-Section A
Question:

If $f(\alpha) = \int_1^{\alpha} \frac{\log_{10} t}{1+t}\,dt$, $\alpha > 0$, then $f(e^3) + f(e^{-3})$ is equal to :

A 9

B $\frac{9}{2}$

C

$$\frac{9}{\log_e(10)}$$

D $$\frac{9}{2\log_e(10)}$$

Answer Given By Candidate:**Not Attempted**

Question ID:100216
Topic Name: Mathematics-Section A
Question: The area of the region $\{(x, y): |x-1| \le y \le \sqrt{5-x^2}\}$ is equal to :

A $$\frac{5}{2}\sin^{-1}\left(\frac{3}{5}\right) - \frac{1}{2}$$

B $$\frac{5\pi}{4} - \frac{3}{2}$$

C $$\frac{3\pi}{4} + \frac{3}{2}$$

D $$\frac{5\pi}{4} - \frac{1}{2}$$

Answer Given By Candidate:**A**

Question ID:100217
Topic Name: Mathematics-Section A
Question:
Let the focal chord of the parabola P : $y^2=4x$ along the line L : $y=mx+c$, m > 0 meet the parabola at the points M and N. Let the line L be a tangent to the hyperbola H : $x^2-y^2=4$. If O is the vertex of P and F is the focus of H on the positive x-axis, then the area of the quadrilateral OMFN is :

A $2\sqrt{6}$

B $2\sqrt{14}$

C $4\sqrt{6}$

D $4\sqrt{14}$

Answer Given By Candidate:**Not Attempted**

Question ID:100218
Topic Name: Mathematics-Section A
Question: The number of points, where the function $f: \mathbf{R} \to \mathbf{R}$, $f(x)=|x-1|\cos|x-2|\sin|x-1|+(x-3)|x^2-5x+4|$, is **NOT** differentiable, is :

A 1

B 2

C 3

D 4

Answer Given By Candidate:**B**

Question ID:100219
Topic Name: Mathematics-Section A
Question:

Let S = {1, 2, 3, ..., 2022}. Then the probability, that a randomly chosen number n from the set S such that HCF (n, 2022) = 1, is :

A $\frac{128}{1011}$

B $\frac{166}{1011}$

C $\frac{127}{337}$

D $\frac{112}{337}$

Answer Given By Candidate:**A**

Question ID:100220
Topic Name: Mathematics-Section A

Let $f(x) = 3^{(x^2-2)^3+4}$, $x \in \mathbf{R}$. Then which of the following statements are true ?

P : $x = 0$ is a point of local minima of f

Q : $x = \sqrt{2}$ is a point of inflection of f

R : f' is increasing for $x > \sqrt{2}$

Question:

A Only P and Q

B Only P and R

C Only Q and R

D All P, Q and R

Answer Given By Candidate:**D**

Question ID:100221
Topic Name: Mathematics-Section B
Question:

Let $S = \{\theta \in (0, 2\pi) : 7\cos^2\theta - 3\sin^2\theta - 2\cos^2 2\theta = 2\}$. Then, the sum of roots of all the equations $x^2 - 2(\tan^2\theta + \cot^2\theta)x + 6\sin^2\theta = 0$, $\theta \in S$, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100222
Topic Name: Mathematics-Section B

Question:

Let the mean and the variance of 20 observations $x_1, x_2, \ldots, x_{20}$ be 15 and 9, respectively. For $\alpha \in \mathbf{R}$, if the mean of $(x_1+\alpha)^2, (x_2+\alpha)^2, \ldots, (x_{20}+\alpha)^2$ is 178, then the square of the maximum value of α is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100223
Topic Name: Mathematics-Section B
Question:

Let a line with direction ratios a, $-4a$, -7 be perpendicular to the lines with direction ratios 3, -1, 2b and b, a, -2. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane $x-y+z=0$ is (α, β, γ), then $\alpha+\beta+\gamma$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100224
Topic Name: Mathematics-Section B
Question:

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\sum_{r=1}^{\infty} \frac{a_r}{2^r} = 4$, then $4a_2$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100225
Topic Name: Mathematics-Section B
Question:

Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6} : 1$. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100226
Topic Name: Mathematics-Section B
Question:

The number of matrices of order 3×3, whose entries are either 0 or 1 and the sum of all the entries is a prime number, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100227
Topic Name: Mathematics-Section B
Question:

Let p and $p+2$ be prime numbers and let

$$\Delta = \begin{vmatrix} p! & (p+1)! & (p+2)! \\ (p+1)! & (p+2)! & (p+3)! \\ (p+2)! & (p+3)! & (p+4)! \end{vmatrix}$$

Then the sum of the maximum values of α and β, such that p^α and $(p+2)^\beta$ divide Δ, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100228
Topic Name: Mathematics-Section B
Question:

If $\frac{1}{2\times3\times4}+\frac{1}{3\times4\times5}+\frac{1}{4\times5\times6}+\ldots+\frac{1}{100\times101\times102}=\frac{k}{101}$, then 34 k is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100229
Topic Name: Mathematics-Section B
Question:

Let $S=\{4, 6, 9\}$ and $T=\{9, 10, 11, ..., 1000\}$. If $A=\{a_1+a_2+...+a_k : k \in \mathbf{N}, a_1, a_2, a_3, ..., a_k \in S\}$, then the sum of all the elements in the set $T-A$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100230
Topic Name: Mathematics-Section B
Question:

Let the mirror image of a circle $c_1 : x^2+y^2-2x-6y+\alpha=0$ in line $y=x+1$ be $c_2 : 5x^2+5y^2+10gx+10fy+38=0$. If r is the radius of circle c_2, then $\alpha+6r^2$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100231
Topic Name: Physics-Section A
Question:

Given below are two statements : One is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.

Assertion (A) : Time period of oscillation of a liquid drop depends on surface tension (S), if density of the liquid is ρ and radius of the drop is r, then $T=K\sqrt{\rho r^3/S^{3/2}}$ is dimensionally correct, where K is dimensionless.

Reason (R) : Using dimensional analysis we get R.H.S. having different dimension than that of time period.

In the light of above statements, choose the correct answer from the options given below.

A Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

B Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

C **(A)** is true but **(R)** is false

D **(A)** is false but **(R)** is true

Answer Given By Candidate:**D**

Question ID:100232
Topic Name: Physics-Section A

Question:

A ball is thrown up vertically with a certain velocity so that, it reaches a maximum height h.

Find the ratio of the times in which it is at height $\frac{h}{3}$ while going up and coming down respectively.

A $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

B $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

C $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

D $\frac{1}{3}$

Answer Given By Candidate: **Not Attempted**

Question ID: 100233
Topic Name: Physics-Section A

Question:

If $t = \sqrt{x} + 4$, then $\left(\frac{dx}{dt}\right)_{t=4}$ is :

A 4

B zero

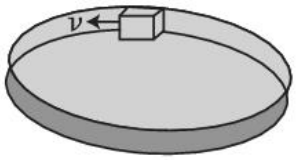
C 8

D 16

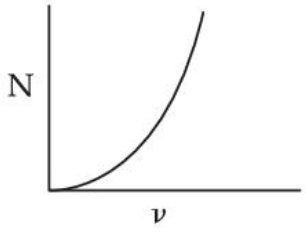
Answer Given By Candidate: **B**

Question ID: 100234
Topic Name: Physics-Section A
Question:

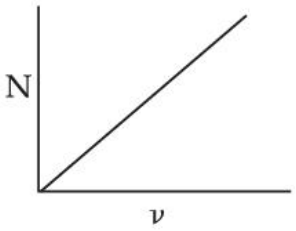
A smooth circular groove has a smooth vertical wall as shown in figure. A block of mass m moves against the wall with a speed v. Which of the following curve represents the correct relation between the normal reaction on the block by the wall (N) and speed of the block (v) ?



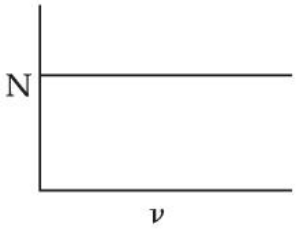
A



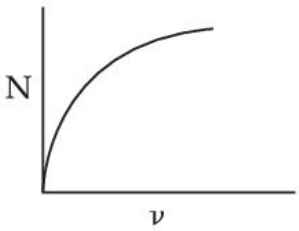
B



C



D



Answer Given By Candidate:**A**

Question ID:100235
Topic Name: Physics-Section A
Question:

A ball is projected with kinetic energy E, at an angle of 60° to the horizontal. The kinetic energy of this ball at the highest point of its flight will become :

A Zero

B $\frac{E}{2}$

C $\frac{E}{4}$

D E

Answer Given By Candidate:**C**

Question ID:100236
Topic Name: Physics-Section A
Question:

Two bodies of mass 1 kg and 3 kg have position vectors $\hat{i}+2\hat{j}+\hat{k}$ and $-3\hat{i}-2\hat{j}+\hat{k}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector :

A $\hat{i}+2\hat{j}+\hat{k}$

B $-3\hat{i}-2\hat{j}+\hat{k}$

C

$-2\hat{i}+2\hat{k}$

D $-2\hat{i}-\hat{j}+2\hat{k}$

Answer Given By Candidate:**A**

Question ID:100237
Topic Name: Physics-Section A
Question:

Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R) : Because the angle of contact between the oil/grease and water is obtuse.

In the light of the above statements, choose the correct answer from the option given below.

A Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

B Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

C **(A)** is true but **(R)** is false

D **(A)** is false but **(R)** is true

Answer Given By Candidate:**A**

Question ID:100238
Topic Name: Physics-Section A
Question:

If the length of a wire is made double and radius is halved of its respective values. Then, the Young's modulus of the material of the wire will :

A remain same

B become 8 times its initial value

C become $\frac{1}{4}^{\text{th}}$ of its initial value

D become 4 times its initial value

Answer Given By Candidate:**B**

Question ID:100239
Topic Name: Physics-Section A
Question:

The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α, is given by :

A $2\pi\sqrt{L/(g\cos\alpha)}$

B $2\pi\sqrt{L/(g\sin\alpha)}$

C $2\pi\sqrt{L/g}$

D $2\pi\sqrt{L/(g\tan\alpha)}$

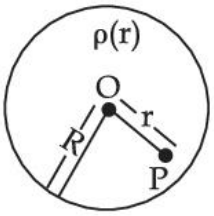
Answer Given By Candidate:**A**

Question ID:100240
Topic Name: Physics-Section A
Question:

A spherically symmetric charge distribution is considered with charge density varying as

$$\rho(r) = \begin{cases} \rho_0\left(\frac{3}{4} - \frac{r}{R}\right) & \text{for } r \le R \\ \text{zero} & \text{for } r > R \end{cases}$$

Where, r(r < R) is the distance from the centre O (as shown in figure). The electric field at point P will be :



A $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{3}{4} - \frac{r}{R}\right)$

B $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{3}{4} - \frac{r}{R}\right)$

C $\frac{\rho_0 r}{4\varepsilon_0}\left(1 - \frac{r}{R}\right)$

D $\frac{\rho_0 r}{5\varepsilon_0}\left(1 - \frac{r}{R}\right)$

Answer Given By Candidate:**B**

Question ID:100241
Topic Name: Physics-Section A
Question:

Given below are two statements.

Statement I : Electric potential is constant within and at the surface of each conductor.

Statement II : Electric field just outside a charged conductor is perpendicular to the surface of the conductor at every point.

In the light of the above statements, choose the most appropriate answer from the options given below.

A Both **statement I** and **statement II** are correct

B

Both **statement I** and **statement II** are incorrect

C **Statement I** is correct but **statement II** is incorrect

D **Statement I** is incorrect but **statement II** is correct

Answer Given By Candidate:**A**

Question ID:100242
Topic Name: Physics-Section A
Question:

Two metallic wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the these wires respectively, the effective conductivity of the combination is :

A $\frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$

B $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$

C $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$

D $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

Answer Given By Candidate:**D**

Question ID:100243
Topic Name: Physics-Section A
Question:

An alternating emf $E = 440 \sin 100\pi t$ is applied to a circuit containing an inductance of $\frac{\sqrt{2}}{\pi}$ H.

If an a.c. ammeter is connected in the circuit, its reading will be :

A 4.4 A

B 1.55 A

C 2.2 A

D 3.11 A

Answer Given By Candidate:**A**

Question ID:100244
Topic Name: Physics-Section A

Question:

A coil of inductance 1 H and resistance 100 Ω is connected to a battery of 6 V. Determine approximately :

(a) The time elapsed before the current acquires half of its steady - state value.

(b) The energy stored in the magnetic field associated with the coil at an instant 15 ms after the circuit is switched on. (Given $\ln 2 = 0.693$, $e^{-3/2} = 0.25$)

A t = 10 ms; U = 2 mJ

B t = 10 ms; U = 1 mJ

C t = 7 ms; U = 1 mJ

D t = 7 ms; U = 2 mJ

Answer Given By Candidate: **Not Attempted**

Question ID: 100245
Topic Name: Physics-Section A

Match **List - I** with **List - II** :

	List - I		**List - II**
(a)	UV rays	(i)	Diagnostic tool in medicine
(b)	X-rays	(ii)	Water purification
(c)	Microwave	(iii)	Communication, Radar
(d)	Infrared wave	(iv)	Improving visibility in foggy days

Choose the correct answer from the options given below :

Question:

A (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

B (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

C (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

D (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

Answer Given By Candidate: **B**

Question ID: 100246
Topic Name: Physics-Section A
Question:

The kinetic energy of emitted electron is E when the light incident on the metal has wavelength λ. To double the kinetic energy, the incident light must have wavelength :

A $\frac{hc}{E\lambda - hc}$

B $\frac{hc\lambda}{E\lambda + hc}$

C

$\frac{h\lambda}{E\lambda + hc}$

D $\frac{hc\lambda}{E\lambda - hc}$

Answer Given By Candidate:**B**

Question ID:100247
Topic Name: Physics-Section A
Question:

Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level.

A 3 : 4

B 4 : 3

C 1 : 4

D 4 : 1

Answer Given By Candidate:**A**

Question ID:100248
Topic Name: Physics-Section A
Question:

Find the modulation index of an AM wave having 8 V variation where maximum amplitude of the AM wave is 9 V.

A 0.8

B 0.5

C 0.2

D 0.1

Answer Given By Candidate:**Not Attempted**

Question ID:100249
Topic Name: Physics-Section A
Question:

A travelling microscope has 20 divisions per cm on the main scale while its vernier scale has total 50 divisions and 25 vernier scale divisions are equal to 24 main scale divisions, what is the least count of the travelling microscope ?

A 0.001 cm

B 0.002 mm

C 0.002 cm

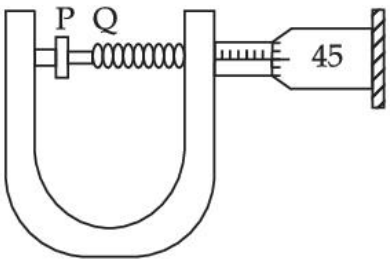
D 0.005 cm

Answer Given By Candidate:**Not Attempted**

Question ID:100250

Topic Name: Physics-Section A
Question:

In an experiment to find out the diameter of wire using screw gauge, the following observations were noted :



(A) Screw moves 0.5 mm on main scale in one complete rotation

(B) Total divisions on circular scale = 50

(C) Main scale reading is 2.5 mm

(D) 45th division of circular scale is in the pitch line

(E) Instrument has 0.03 mm negative error

Then the diameter of wire is :

A 2.92 mm

B 2.54 mm

C 2.98 mm

D 3.45 mm

Answer Given By Candidate:**Not Attempted**

Question ID:100251
Topic Name: Physics-Section B
Question:

An object is projected in the air with initial velocity u at an angle θ. The projectile motion is such that the horizontal range R, is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the case. The value of the angle of projection, at which the second object is projected, will be _____ degree.

Answer Given By Candidate:**15**

Question ID:100252
Topic Name: Physics-Section B
Question:

If the acceleration due to gravity experienced by a point mass at a height h above the surface of earth is same as that of the acceleration due to gravity at a depth αh ($h << R_e$) from the earth surface. The value of α will be _____.

(use $R_e = 6400$ km)

Answer Given By Candidate:**Not Attempted**

Question ID:100253
Topic Name: Physics-Section B
Question:

The pressure P_1 and density d_1 of diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes suddenly to $P_2(>P_1)$ and d_2 respectively during an adiabatic process. The temperature of the gas increases and becomes _____ times of its initial temperature. (given $\frac{d_2}{d_1} = 32$)

Answer Given By Candidate:**Not Attempted**

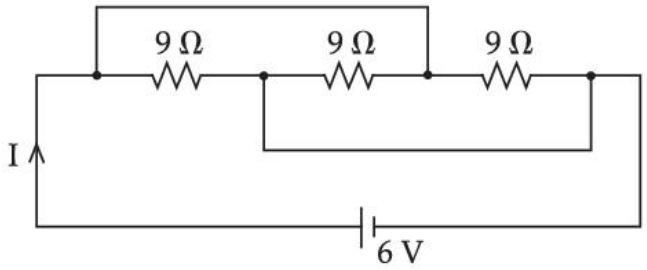
Question ID:100254
Topic Name: Physics-Section B
Question:

One mole of a monoatomic gas is mixed with three moles of a diatomic gas. The molecular specific heat of mixture at constant volume is $\frac{\alpha^2}{4}$R J/mol K; then the value of α will be _____. (Assume that the given diatomic gas has no vibrational mode).

Answer Given By Candidate:**Not Attempted**

Question ID:100255
Topic Name: Physics-Section B

The current I flowing through the given circuit will be _____ A.



Question:

Answer Given By Candidate:**1**

Question ID:100256
Topic Name: Physics-Section B
Question:

A closely wounded circular coil of radius 5 cm produces a magnetic field of 37.68×10^{-4} T at its center. The current through the coil is _____ A.

[Given, number of turns in the coil is 100 and $\pi = 3.14$]

Answer Given By Candidate:**2**

Question ID:100257
Topic Name: Physics-Section B

Question:

Two light beams of intensities 4I and 9I interfere on a screen. The phase difference between these beams on the screen at point A is zero and at point B is π. The difference of resultant intensities, at the point A and B, will be _____ I.

Answer Given By Candidate:**Not Attempted**

Question ID:100258
Topic Name: Physics-Section B
Question:

A wire of length 314 cm carrying current of 14 A is bent to form a circle. The magnetic moment of the coil is _____ A-m^2. [Given $\pi = 3.14$]

Answer Given By Candidate:**11**

Question ID:100259
Topic Name: Physics-Section B
Question:

The X-Y plane be taken as the boundary between two transparent media M_1 and M_2. M_1 in $Z \geq 0$ has a refractive index of $\sqrt{2}$ and M_2 with $Z < 0$ has a refractive index of $\sqrt{3}$. A ray of light travelling in M_1 along the direction given by the vector $\vec{P} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$, is incident on the plane of separation. The value of difference between the angle of incident in M_1 and the angle of refraction in M_2 will be _____ degree.

Answer Given By Candidate:**Not Attempted**

Question ID:100260
Topic Name: Physics-Section B
Question:

If the potential barrier across a p-n junction is 0.6 V. Then the electric field intensity, in the depletion region having the width of 6×10^{-6} m, will be _____ $\times 10^5$ N/C.

Answer Given By Candidate:**Not Attempted**

Question ID:100261
Topic Name: Chemistry-Section A
Question:

Which of the following pair of molecules contain odd electron molecule and an expanded octet molecule ?

A BCl_3 and SF_6

B NO and H_2SO_4

C SF_6 and H_2SO_4

D BCl_3 and NO

Answer Given By Candidate:**C**

Question ID:100262
Topic Name: Chemistry-Section A

Question:

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

20 g 5 g

Consider the above reaction, the limiting reagent of the reaction and number of moles of NH_3 formed respectivly are :

A H_2, 1.42 moles

B H_2, 0.71 moles

C N_2, 1.42 moles

D N_2, 0.71 moles

Answer Given By Candidate: **B**

Question ID: 100263
Topic Name: Chemistry-Section A
Question:

100 mL of 5% (w/v) solution of NaCl in water was prepared in 250 mL beaker. Albumin from the egg was poured into NaCl solution and stirred well. This resulted in a/an :

A Lyophilic sol

B Lyophobic sol

C Emulsion

D Precipitate

Answer Given By Candidate: **C**

Question ID: 100264
Topic Name: Chemistry-Section A
Question:

The first ionization enthalpy of Na, Mg and Si, respectively, are : 496, 737 and 786 kJ mol^{-1}. The first ionization enthalpy (kJ mol^{-1}) of Al is :

A 487

B 768

C 577

D 856

Answer Given By Candidate: **B**

Question ID: 100265
Topic Name: Chemistry-Section A
Question: In metallurgy the term "gangue" is used for :

A Contamination of undesired earthy materials.

B Contamination of metals, other than desired metal.

C Minerals which are naturally occuring in pure form

D Magnetic impurities in an ore.

Answer Given By Candidate:**A**

Question ID:100266
Topic Name: Chemistry-Section A
Question: The reaction of zinc with excess of aqueous alkali, evolves hydrogen gas and gives :

A $Zn(OH)_2$

B ZnO

C $[Zn(OH)_4]^{2-}$

D $[ZnO_2]^{2-}$

Answer Given By Candidate:**B**

Question ID:100267
Topic Name: Chemistry-Section A
Question: Lithium nitrate and sodium nitrate, when heated separately, respectively, give :

A $LiNO_2$ and $NaNO_2$

B Li_2O and Na_2O

C Li_2O and $NaNO_2$

D $LiNO_2$ and Na_2O

Answer Given By Candidate:**C**

Question ID:100268
Topic Name: Chemistry-Section A
Question:
Number of lone pairs of electrons in the central atom of SCl_2, O_3, ClF_3 and SF_6, respectively, are :

A 0, 1, 2 and 2

B 2, 1, 2 and 0

C 1, 2, 2 and 0

D 2, 1, 0 and 2

Answer Given By Candidate:**B**

Question ID:100269
Topic Name: Chemistry-Section A
Question: In following pairs, the one in which both transition metal ions are colourless is :

A Sc^{3+}, Zn^{2+}

B Ti^{4+}, Cu^{2+}

C V^{2+}, Ti^{3+}

D Zn^{2+}, Mn^{2+}

Answer Given By Candidate:**A**

Question ID:100270
Topic Name: Chemistry-Section A
Question:

In neutral or faintly alkaline medium, $KMnO_4$ being a powerful oxidant can oxidize, thiosulphate almost quantitatively, to sulphate. In this reaction overall change in oxidation state of manganese will be :

A 5

B 1

C 0

D 3

Answer Given By Candidate:**D**

Question ID:100271
Topic Name: Chemistry-Section A
Question: Which among the following pairs has only herbicides ?

A Aldrin and Dieldrin

B Sodium chlorate and Aldrin

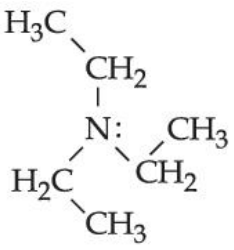
C Sodium arsinate and Dieldrin

D Sodium chlorate and sodium arsinite.

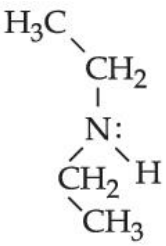
Answer Given By Candidate:**Not Attempted**

Question ID:100272
Topic Name: Chemistry-Section A
Question: Which among the following is the strongest Bronsted base ?

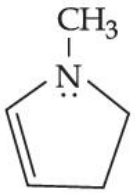
A



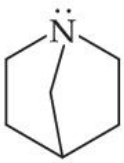
B



C



D

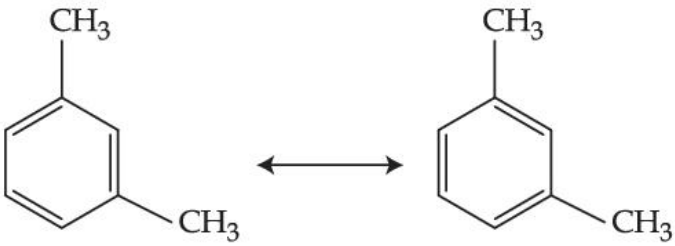


Answer Given By Candidate:**A**

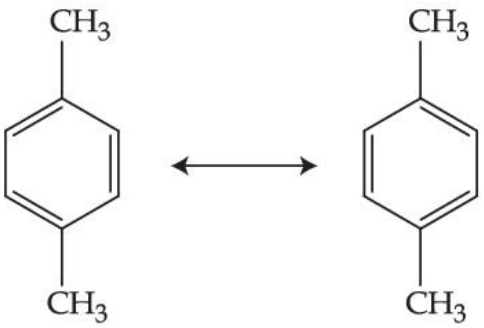
Question ID:100273
Topic Name: Chemistry-Section A
Question:

Which among the following pairs of the structures will give different products on ozonolysis ? (Consider the double bonds in the structures are rigid and not delocalized.)

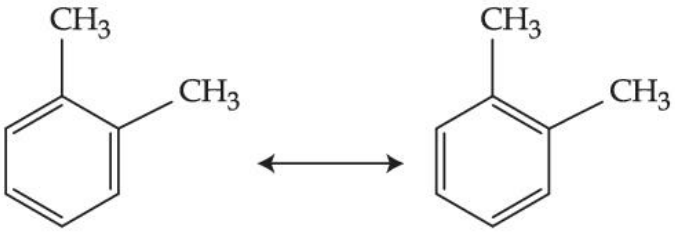
A



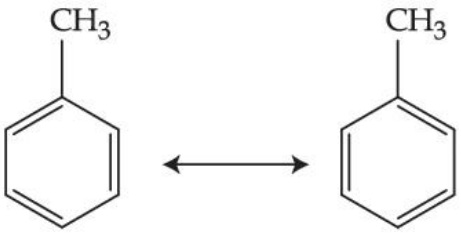
B



C



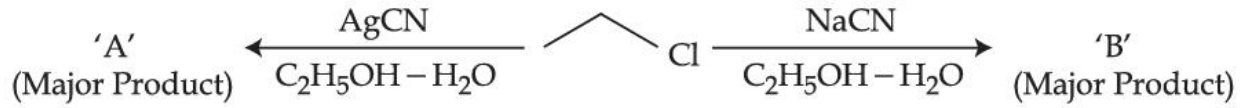
D



Answer Given By Candidate:**C**

Question ID:100274
Topic Name: Chemistry-Section A

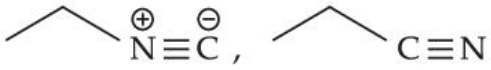
Question:



Considering the above reactions, the compound 'A' and compound 'B' respectively are :

A 

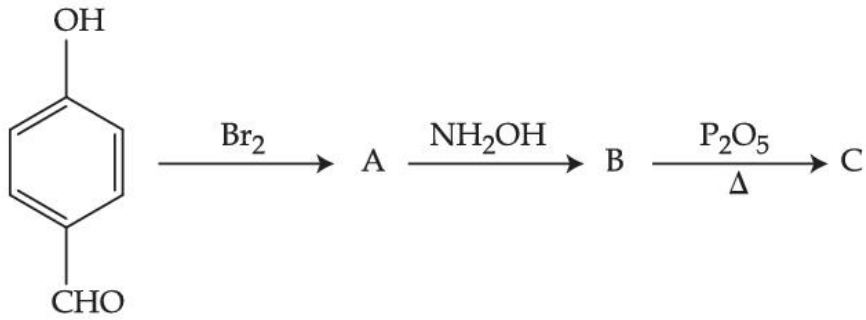
B 

C 

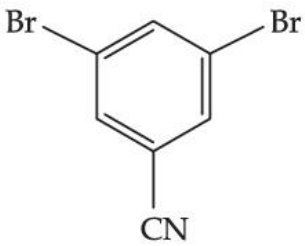
D 

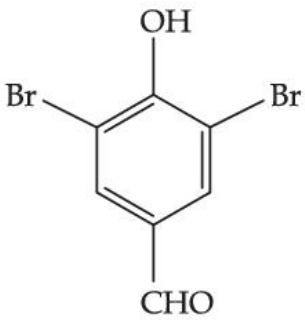
Answer Given By Candidate: **C**

Question ID: 100275
Topic Name: Chemistry-Section A

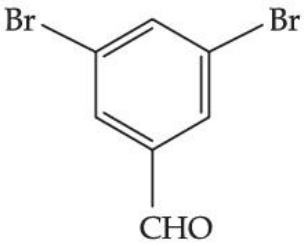


Question: Consider the above reaction sequence, the Product 'C' is :

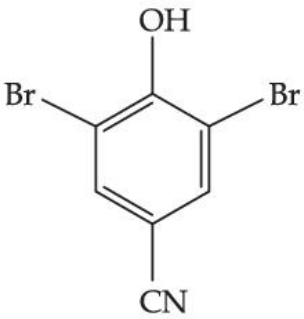
A 

B 

C

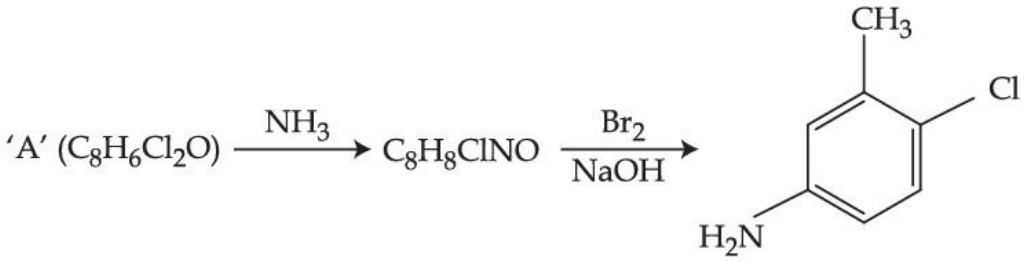


D



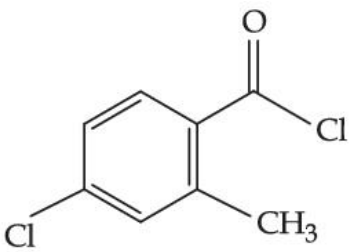
Answer Given By Candidate:**A**

Question ID:100276
Topic Name: Chemistry-Section A

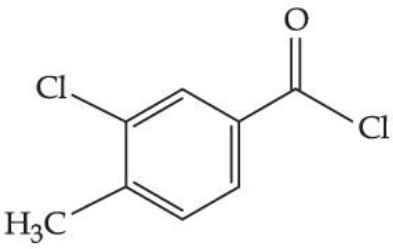


Question: Consider the above reaction, the compound 'A' is :

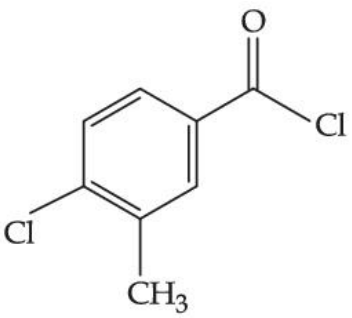
A



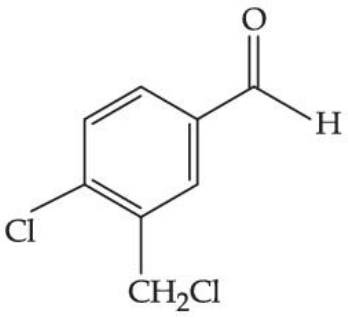
B



C

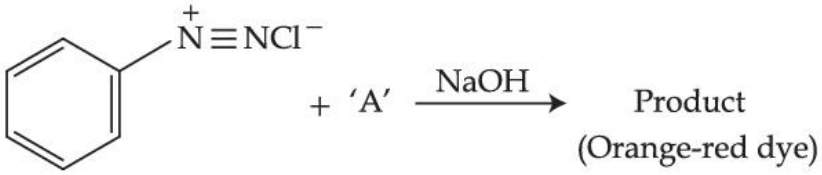


D



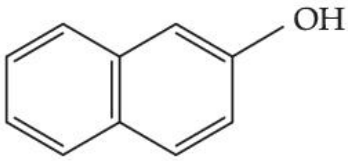
Answer Given By Candidate: **C**

Question ID: 100277
Topic Name: Chemistry-Section A

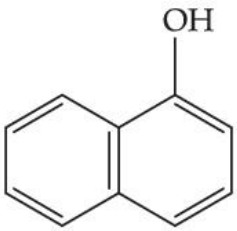


Question: Which among the following represent reagent 'A' ?

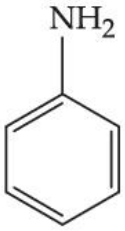
A



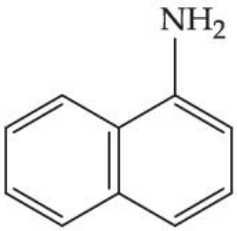
B



C



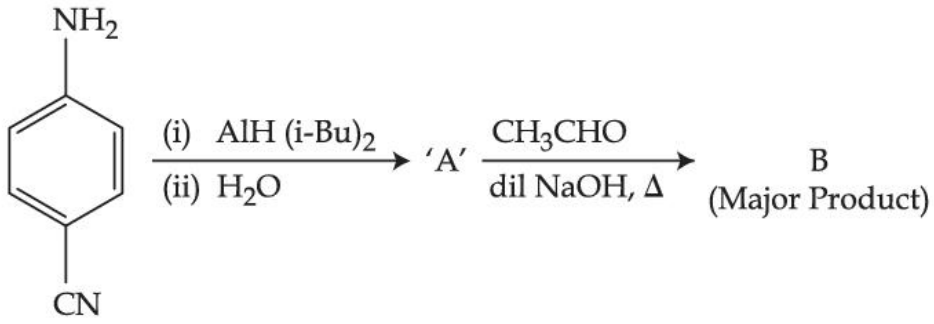
D



Answer Given By Candidate: **A**

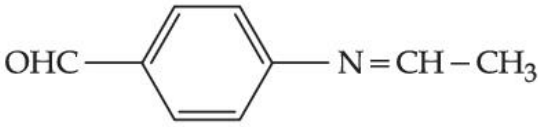
Question ID: 100278
Topic Name: Chemistry-Section A

Consider the following reaction sequence :

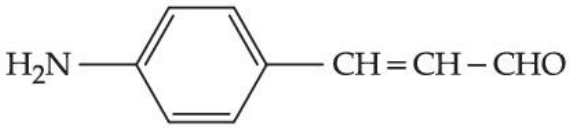


Question: The product 'B' is :

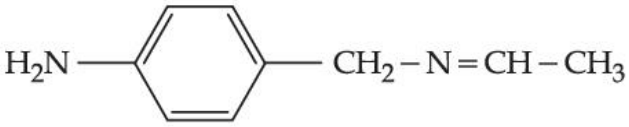
A



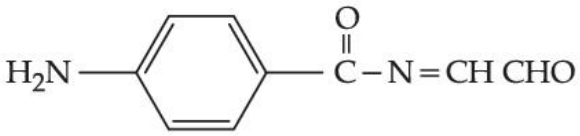
B



C



D



Answer Given By Candidate: **B**

Question ID: 100279
Topic Name: Chemistry-Section A
Question: Which of the following compounds is an example of hypnotic drug ?

A Seldane

B Amytal

C Aspartame

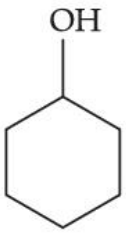
D Prontosil

Answer Given By Candidate: **Not Attempted**

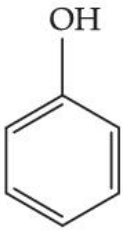
Question ID: 100280
Topic Name: Chemistry-Section A
Question:

A compound 'X' is acidic and it is soluble in NaOH solution, but insoluble in $NaHCO_3$ solution. Compound 'X' also gives violet colour with neutral $FeCl_3$ solution. The compound 'X' is :

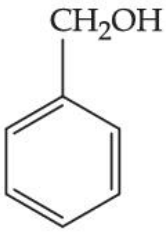
A



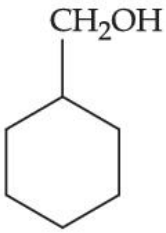
B



C



D



Answer Given By Candidate:**B**

Question ID:100281
Topic Name: Chemistry-Section B
Question:

Resistance of a conductivity cell (cell constant 129 m^{-1}) filled with 74.5 ppm solution of KCl is 100 Ω (labelled as solution 1). When the same cell is filled with KCl solution of 149 ppm, the resistance is 50 Ω (labelled as solution 2). The ratio of molar conductivity of solution 1 and solution 2 is i.e. $\frac{\wedge_1}{\wedge_2} = x \times 10^{-3}$. The value of x is _____. (Nearest integer)

Given, molar mass of KCl is 74.5 g mol^{-1}.

Answer Given By Candidate:**2**

Question ID:100282
Topic Name: Chemistry-Section B
Question:

Ionic radii of cation A^+ and anion B^- are 102 and 181 pm respectively. These ions are allowed to crystallize into an ionic solid. This crystal has cubic close packing for B^-. A^+ is present in all octahedral voids. The edge length of the unit cell of the crystal AB is _____pm. (Nearest Integer)

Answer Given By Candidate:**Not Attempted**

Question ID:100283
Topic Name: Chemistry-Section B
Question:

The minimum uncertainty in the speed of an electron in an one dimensional region of length $2a_0$ (Where a_0 = Bohr radius 52.9 pm) is _____ km s^{-1}.

(Given : Mass of electron = 9.1×10^{-31} kg, Planck's constant h = 6.63×10^{-34} Js)

Answer Given By Candidate:**Not Attempted**

Question ID:100284
Topic Name: Chemistry-Section B
Question:

When 600 mL of 0.2 M HNO_3 is mixed with 400 mL of 0.1 M NaOH solution in a flask, the rise in temperature of the flask is _____ $\times 10^{-2}$ °C.

(Enthalpy of neutralisation = 57 kJ mol^{-1} and Specific heat of water = 4.2 JK^{-1} g^{-1})

(Neglect heat capacity of flask)

Answer Given By Candidate:**Not Attempted**

Question ID:100285
Topic Name: Chemistry-Section B
Question:

If O_2 gas is bubbled through water at 303 K, the number of millimoles of O_2 gas that dissolve in 1 litre of water is _____. (Nearest Integer)

(Given : Henry's Law constant for O_2 at 303 K is 46.82 k bar and partial pressure of O_2 = 0.920 bar)

(Assume solubility of O_2 in water is too small, nearly negligible)

Answer Given By Candidate:**Not Attempted**

Question ID:100286
Topic Name: Chemistry-Section B
Question:

If the solubility product of PbS is 8×10^{-28}, then the solubility of PbS in pure water at 298 K is $x \times 10^{-16}$ mol L^{-1}. The value of x is _____. (Nearest Integer)

[Given : $\sqrt{2} = 1.41$]

Answer Given By Candidate:**Not Attempted**

Question ID:100287
Topic Name: Chemistry-Section B

Question:

The reaction between X and Y is first order with respect to X and zero order with respect to Y.

Experiment	[X] / mol L^{-1}	[Y] / mol L^{-1}	Initial rate / mol L^{-1} min^{-1}
I	0.1	0.1	2×10^{-3}
II	L	0.2	4×10^{-3}
III	0.4	0.4	$M\times10^{-3}$
IV	0.1	0.2	2×10^{-3}

Examine the data of table and calculate ratio of numerical values of M and L. (Nearest Integer)

Answer Given By Candidate:**80**

Question ID:100288
Topic Name: Chemistry-Section B
Question:

In a linear tetrapeptide (Constituted with different amino acids), (number of amino acids) − (number of peptide bonds) is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100289
Topic Name: Chemistry-Section B
Question:

In bromination of Propyne, with Bromine 1, 1, 2, 2-tetrabromopropane is obtained in 27% yield. The amount of 1, 1, 2, 2-tetrabromopropane obtained from 1 g of Bromine in this reaction is _____ $\times10^{-1}$ g. (Nearest integer)

(Molar Mass : Bromine = 80 g/mol)

Answer Given By Candidate:**Not Attempted**

Question ID:100290
Topic Name: Chemistry-Section B
Question:

$[Fe(CN)_6]^{3-}$ should be an inner orbital complex. Ignoring the pairing energy, the value of crystal field stabilization energy for this complex is (−) _____ Δ_o. (Nearest integer)

Answer Given By Candidate:**Not Attempted**